The Bayesian Lasso

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1 Introduction

Lasso problem

$$\min_{\beta}(y - X\beta)^T(y - X\beta) + \lambda \sum_{j-1}^{p} |\beta_j|$$

where $\lambda \geq 0$ is a tuning parameter.

We consider a fully Bayesian analysis using a conditional Laplace prior specification of the form

$$\pi(\beta|\boldsymbol{\sigma^2}) = \prod_{i=1}^{p} \frac{\lambda}{2\sqrt{\sigma^2}} e^{-\lambda|\beta_j|/\sqrt{\sigma^2}} \tag{1}$$

2 Hierarchical Model and Gibbs Sampler

The Gibbs sampler for the Bayesian Lasso exploits the following representation of the Laplace distribution as a scale mixture of norms(with an exponential mixing density)

$$\frac{a}{2}e^{-a|z|} = \int_0^\infty \frac{1}{\sqrt{2\pi s}} \exp\left(-\frac{z^2}{2s}\right) \frac{a^2}{2} e^{-a^2 s/2}\, ds, \quad a > 0 \tag{2}$$

This suggests the following hierarchical representation of the full model:

$$\begin{aligned} y \mid \mu, X, \beta, \sigma^2 &\sim N_n(\mu 1_n + X\beta, \sigma^2 I_n), \\ \beta \mid \sigma^2, \tau_1^2, \ldots, \tau_p^2 &\sim N_p(0_p, \sigma^2 D_\tau), \\ D_\tau &= \operatorname{diag}(\tau_1^2, \ldots, \tau_p^2), \\ \sigma^2, \tau_1^2, \ldots, \tau_p^2 &\sim \pi(\sigma^2)\, d\sigma^2 \prod_{j=1}^{p} \frac{\lambda^2}{2} e^{-\lambda^2 \tau_j^2/2}\, d\tau_j^2, \\ \sigma^2, \tau_1^2, \ldots, \tau_p^2 &> 0. \end{aligned} \tag{3}$$

Why integrating out $\tau_1^2 ... \tau_p^2$,the conditional prior on β has the desired form (1)?

From (3) we know

$$\beta \mid \sigma^2, \tau_1^2, \ldots, \tau_p^2 \sim N_p(0_p, \sigma^2 D_\tau)$$

Hence

$$\beta_j|\sigma^2, \tau_j^2 \sim N(0, \sigma^2\tau_j^2)$$

Our goal is

$$p(\beta_j|\sigma^2) = \int_0^\infty p(\beta_j|\sigma^2, \tau_j^2) p(\tau_j^2) d\tau_j^2$$

Normal term

$$p(\beta_j|\sigma^2, \tau_j^2) = \frac{1}{\sqrt{2\pi\sigma^2\tau_j^2}} \exp -\frac{\beta_j^2}{2\sigma^2\tau_j^2} \quad \textbf{and} \quad p(\tau_j^2) = \frac{\lambda^2}{2} e^{\frac{-\lambda^2}{2}\tau_j^2}$$

Combine and integrating

$$p(\beta_j \mid \sigma^2) = \int_0^\infty \frac{1}{\sqrt{2\pi\sigma^2\tau_j^2}} \exp\left(-\frac{\beta_j^2}{2\sigma^2\tau_j^2} - \frac{\lambda^2}{2}\tau_j^2\right) d\tau_j^2$$

From the equation (2)

$$a = \frac{\beta_j^2}{2\sigma^2}, \qquad b = \frac{\lambda^2}{2}$$

Substituting gives:

$$2\sqrt{ab} = 2\sqrt{\frac{\beta_j^2}{2\sigma^2} \cdot \frac{\lambda^2}{2}} = \frac{\lambda|\beta_j|}{\sigma}$$

Hence

$$p(\beta_j \mid \sigma^2) \propto \exp\left(-\frac{\lambda|\beta_j|}{\sigma}\right)$$

Full conditional for β

Assume the centered model:

$$\tilde{y} \mid \beta, \sigma^2 \sim N_n(X\beta, \sigma^2 I_n),$$

and the prior:

$$\beta \mid \sigma^2, \tau^2 \sim N_p(0, \sigma^2 D_\tau), \quad D_\tau = \mathrm{diag}(\tau_1^2, \ldots, \tau_p^2),$$

with prior for σ^2 given by

$$\pi(\sigma^2) \propto \frac{1}{\sigma^2}.$$

The conditional posterior of β is proportional to

$$p(\beta \mid \tilde{y}, \sigma^2, \tau^2) \propto p(\tilde{y} \mid \beta, \sigma^2)\, p(\beta \mid \sigma^2, \tau^2).$$

The likelihood is

$$p(\tilde{y} \mid \beta, \sigma^2) \propto (\sigma^2)^{-n/2} \exp\left\{-\frac{1}{2\sigma^2}(\tilde{y} - X\beta)^T(\tilde{y} - X\beta)\right\}.$$

The prior is

$$p(\beta \mid \sigma^2, \tau^2) \propto (\sigma^2)^{-p/2} \exp\left\{-\frac{1}{2\sigma^2}\beta^T D_\tau^{-1}\beta\right\}.$$

Combining the two terms:

$$p(\beta \mid \cdot) \propto \exp\left\{-\frac{1}{2\sigma^2}\left[(\tilde{y} - X\beta)^T(\tilde{y} - X\beta) + \beta^T D_\tau^{-1}\beta\right]\right\}.$$

Expanding the quadratic form:

$$(\tilde{y} - X\beta)^T(\tilde{y} - X\beta) = \tilde{y}^T\tilde{y} - 2\beta^T X^T\tilde{y} + \beta^T X^T X\beta.$$

Thus,

$$p(\beta \mid \cdot) \propto \exp\left\{-\frac{1}{2\sigma^2}\left[\beta^T(X^T X + D_\tau^{-1})\beta - 2\beta^T X^T\tilde{y}\right]\right\}.$$

Let

$$A = X^T X + D_\tau^{-1}.$$

Completing the square:

$$\beta^T A\beta - 2\beta^T X^T\tilde{y} = (\beta - A^{-1}X^T\tilde{y})^T A(\beta - A^{-1}X^T\tilde{y}) + \text{constant}.$$

Therefore,

$$\beta \mid \tilde{y}, \sigma^2, \tau^2 \sim N_p\left(A^{-1}X^T\tilde{y}, \sigma^2 A^{-1}\right).$$

Full conditional for σ^2
We have

$$p(\sigma^2 \mid \tilde{y}, \beta, \tau^2) \propto p(\tilde{y} \mid \beta, \sigma^2)\, p(\beta \mid \sigma^2, \tau^2)\, \pi(\sigma^2).$$

Substituting each term:

$$p(\sigma^2 \mid \cdot) \propto (\sigma^2)^{-n/2} (\sigma^2)^{-p/2} (\sigma^2)^{-1} \exp\left\{ -\frac{1}{2\sigma^2} \left[(\tilde{y} - X\beta)^T (\tilde{y} - X\beta) + \beta^T D_\tau^{-1} \beta \right] \right\}.$$

Simplifying:

$$p(\sigma^2 \mid \cdot) \propto (\sigma^2)^{-(n+p)/2-1} \exp\left\{ -\frac{1}{2\sigma^2} \left[(\tilde{y} - X\beta)^T (\tilde{y} - X\beta) + \beta^T D_\tau^{-1} \beta \right] \right\}.$$

This is the kernel of an inverse-gamma distribution. Accounting for the loss of one degree of freedom due to centering, we obtain:

$$\sigma^2 \mid \tilde{y}, \beta, \tau^2 \sim \mathrm{IG}\left(\frac{n-1}{2} + \frac{p}{2}, \frac{(\tilde{y} - X\beta)^T (\tilde{y} - X\beta)}{2} + \frac{\beta^T D_\tau^{-1} \beta}{2} \right).$$

$\tau_1^2 ... \tau_p^2$ are conditionally indepent, with $\frac{1}{\tau_j^2}$ conditionally inverse-Gaussian

$$\frac{1}{\tau_j^2} \mid \beta_j, \sigma^2 \sim \mathrm{IG}\left(\sqrt{\frac{\lambda^2 \sigma^2}{\beta_j^2}},\ \lambda^2 \right)$$

3 Choosing the Bayesian Lasso Parameter

The regularization parameter λ controls the degree of shrinkage in the Bayesian Lasso model. Larger values of λ lead to stronger shrinkage and sparser estimates, while smaller values result in weaker penalization.

Two approaches are proposed for selecting λ:

1. Empirical Bayes Approach

In the empirical Bayes framework, λ is estimated from the data using a marginal maximum likelihood approach. A Monte Carlo EM algorithm is employed, where each iteration consists of:

- Running a Gibbs sampler using the current value of λ,
- Updating λ using the expectation of latent variables τ_j^2.

The update rule is given by:

$$\lambda^{(k)} = \sqrt{\frac{2p}{\sum_{j=1}^p \mathbb{E}_{k-1}[\tau_j^2]}}.$$

In practice, the expectations are approximated by averages from the Gibbs sampler. The resulting estimates of λ typically fluctuate around the optimal value due to sampling variability.

2. Fully Bayesian Approach (Hyperprior)

Alternatively, λ can be treated as a random variable by assigning a prior distribution. A common choice is a gamma prior on λ^2:

$$\lambda^2 \sim \mathrm{Gamma}(r, \delta).$$

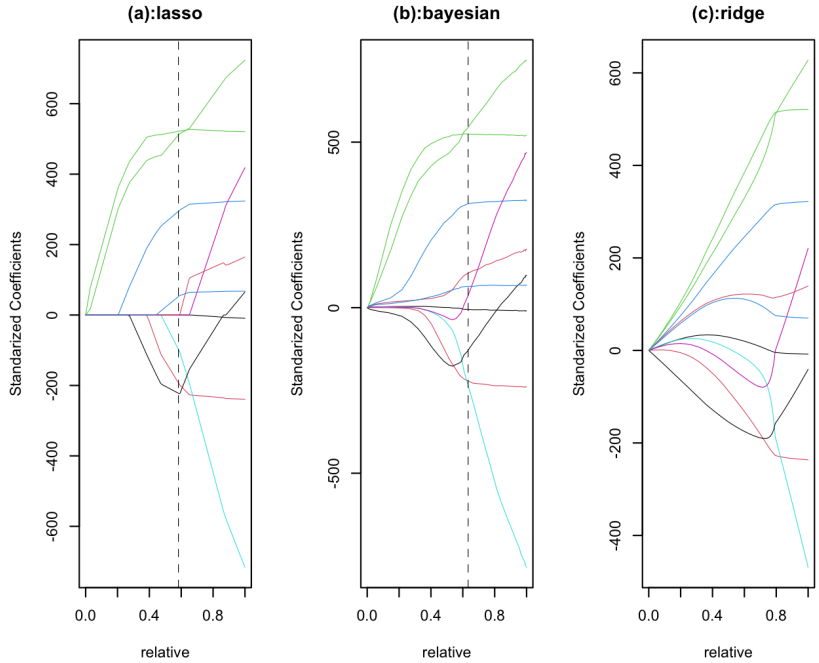
This choice preserves conjugacy, allowing λ^2 to be updated within the Gibbs sampler. The full conditional distribution is:

$$\lambda^2 \mid \cdot \sim \mathrm{Gamma}\left(p + r,\ \sum_{j=1}^p \frac{\tau_j^2}{2} + \delta \right).$$

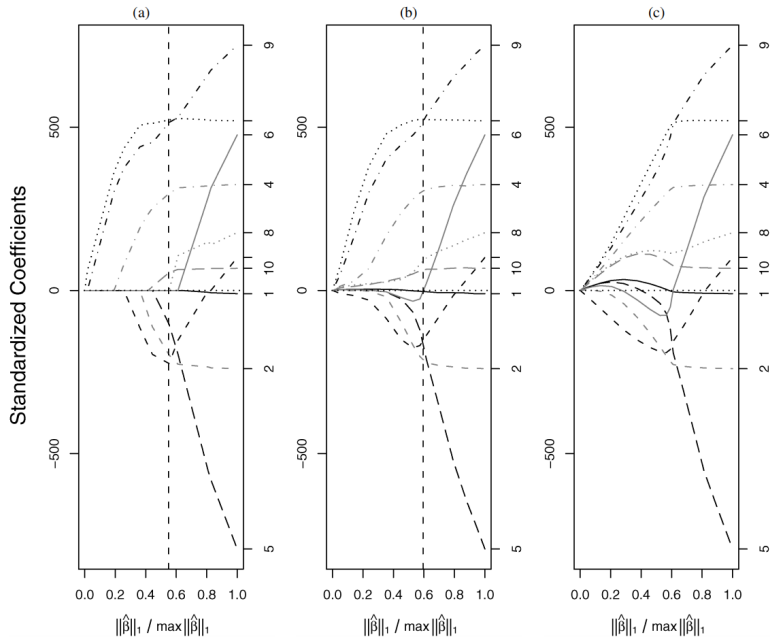
This approach allows uncertainty in λ to be incorporated into posterior inference.

Both methods aim to determine an appropriate level of shrinkage. The empirical Bayes method provides a data-driven point estimate of λ, while the fully Bayesian approach treats λ as a random variable and estimates it jointly with other parameters. In practice, both approaches yield similar results.

4 Reproduction



(a) Reproduce

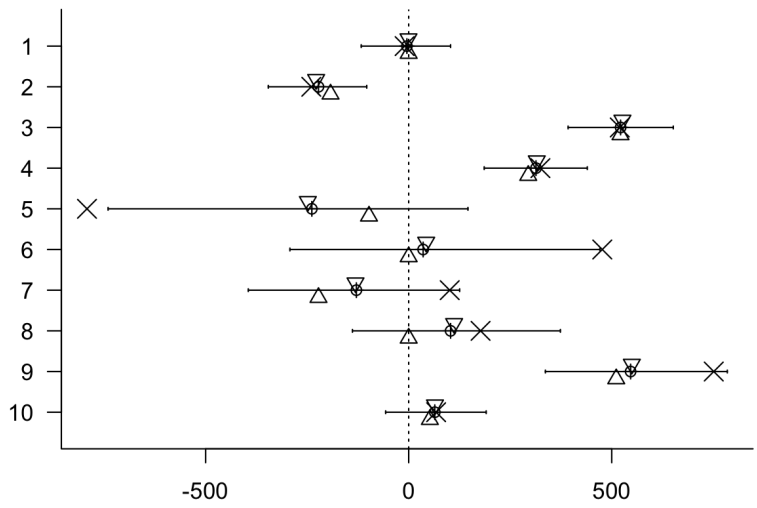


(b) True image

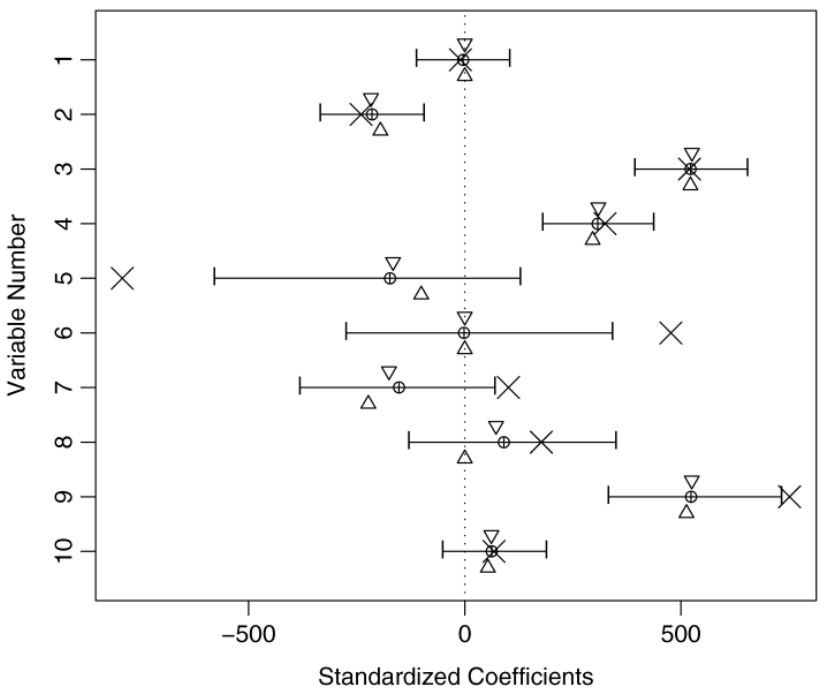
Figure 1

It is very clear that the plot I reproduced is very similar to the plot form paper.And also the λ is .237

Remark: When I reproduce this plot, I met very problems. So the first thing is from the range of the y-axis in the figure, we can see that it starts at zero. This implies that a very large value of λ is used. Therefore, when implementing the code, we need to manually specify an appropriate range for λ.



(a) Reproduce



(b) True image

Figure 2

Same as paper